

Supplementary Materials

TIMBRE: Layer-Wise Cross-Lingual Speech Emotion Recognition Across 49 Layers and 26 Corpora

Anatoly Marchenko

1 Corpus metadata (Table S1)

Table 1 lists, for each of the 26 corpora, the language, elicitation style, number of utterances used (after per-class balancing, max. 500 per class), the number of emotion labels (harmonized to the 4 core emotions *angry*, *happy*, *sad*, *neutral*), and prosodic type. Corpora are ordered by size.

Table 1: Per-corpus metadata. Elicitation: *acted* = scripted/performed; *crowd-sourced* = scripted, performed by crowd workers; *natural* = naturally occurring; Size = utterances after per-class balancing (≤ 500 per class). Labels = number of emotion classes retained. For the three corpora marked with *, the wav2vec2 experiments (Exps. 2–3) used slightly larger pools than the acoustic runs listed here, owing to cache regeneration between experiment phases: CAVES 1975, JNVN 786, Thai-SER 2000. Rhythm-class labels (stress-/syllable-timed) follow traditional impressionistic classifications; the discreteness of rhythm classes is contested (Ramus et al., 1999; Grabe & Low, 2002), and these labels are not used in any quantitative analysis in the paper (the analysis groups are tonal, Romance, Germanic, Slavic).

Corpus	Language	Elicitation	Size	Labels	Prosodic
DUSHA	Russian	crowd-sourced	2000	4	stress-timed
Emozionalmente	Italian	acted	2000	4	syllable-timed
ESD-ZH	Mandarin	acted	2000	4	tonal
KazEmoTTS	Kazakh	acted (read, 3 narrators)	2000	4	syllable-timed
nEMO	Polish	acted	2000	4	stress-timed
PAVOQUE	German	acted	2000	4	stress-timed
SUBESCO	Bengali	acted	2000	4	syllable-timed
ViSEC	Vietnamese	acted	2000	4	tonal
CAVES	Cantonese	acted	1969*	4	tonal
ASED	Amharic	acted	1942	4	stress-timed
Thai-SER	Thai	acted	1914*	4	tonal
TurEV-DB	Turkish	acted	1735	4	syllable-timed
ShEMO	Persian	acted	1511	4	syllable-timed
Estonian-EEKK	Estonian	acted	1164	4	stress-timed
RAVDESS	English	acted	864	4	stress-timed
RESO	Russian	acted	790	4	stress-timed
JNVN	Japanese	acted	602*	4	pitch-accent
MESD	Spanish	acted	574	4	syllable-timed
Hi,KIA	Korean	acted	484	4	syllable-timed
MDER-MA	Arabic-Moroccan	acted	420	4	stress-timed
URDU-Dataset	Urdu	natural	400	4	syllable-timed
emoUERJ	Portuguese-BR	acted	377	4	syllable-timed
EmoDB	German	acted	312	4	stress-timed
Oreau	French	acted	288	4	syllable-timed
CaFE	French	acted	129	4	syllable-timed
Kannada-SER	Kannada	acted	116	4	syllable-timed

2 Statistical detail for Section 4.2

All membership contrasts at L15 use corpus-clustered bootstrap (2,000 resamples, resampling corpora with replacement; seed 20260607). Within- vs. between-group (the four analysis groups, computed over their 15 member corpora; 44 within vs. 166 between pairs): 0.410 vs. 0.425, difference 95% CI $[-0.061, +0.029]$. Within- vs. between-family (restricted to families with at least two corpora; 19 corpora, 40 within vs. 302 between pairs): 0.393 vs. 0.407, difference 95% CI $[-0.071, +0.036]$. Both intervals span zero, indicating no significant membership advantage; the grand average over all 650 off-diagonal pairs at L15 is 0.392.

3 Classifier and preprocessing detail

All cross-corpus probes use scikit-learn `LogisticRegression` (L2, $C=1.0$, `lbfgs`, `max_iter=1000`, `random_state=42`) and report weighted F1. In the layer-wise tensor (Exp. 3), representations are z-scored with `StandardScaler` fitted on the training corpus only; in the acoustic cross-corpus run (Exp. 1), raw feature values are used without normalization. No test-corpus statistics are used in either case. In the per-corpus ablation (Exp. 4), features are z-scored within each training fold. Class balancing subsamples each corpus to at most 500 utterances per emotion before any experiment (sizes in Table S1). The acoustic 26×26 matrix (`cpu_acoustic_cross_26_logreg.json`) is included in the released data alongside the wav2vec2 matrices.